Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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This document reports, in full, four robustness checks summarized in the main text, and the variants of the candidate measure (Section S5). All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption, and Table S6 lists the script behind every result reported in the main text.

Table S1: Panel-size recoverability under alternative correlation bars

The main text calls a panel size "recovering" the full-panel ranking once mean Spearman ρ against the full panel exceeds 0.90, a bar borrowed from the "excellent" range of intraclass-correlation guidelines (Results). Table S1 reports p^* (the smallest panel size reaching the bar) at four bars, including the main-text bar of 0.90, for all four datasets.

The ratio remains the slowest to recover on every dataset for every bar tested, and the Gini coefficient the second-slowest on Davis, Klaeger, and Anastassiadis. On Metz it ties the entropy at every bar. No bar in $[0.80, 0.95]$ reverses the order of two definitions on any dataset. The bar decides only whether some of them tie.

Table S1: Panel-size p^* (kinases) at four correlation bars. (`pstar_bar_sensitivity.py`)

Davis (433 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	80	50	350
bar = 0.85	50	80	50	380
bar = 0.90	50	110	50	410
bar = 0.95	50	170	80	410

Klaeger (343 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	260	50	320
bar = 0.85	50	260	80	320
bar = 0.90	80	290	110	343
bar = 0.95	140	320	200	343

Anastassiadis (300 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	170	50	260
bar = 0.85	50	170	50	260
bar = 0.90	50	200	50	290
bar = 0.95	80	260	80	290

Metz (172 kinases)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets to check whether they behave as additional members of the ratio family (WS, RS) or the distribution-based family (CATDS), as claimed in the Discussion. (`bosc_catds_comparison.py`)

Table S2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.804	0.878	0.509	0.955 / 0.960
Klaeger	222	0.909	0.892	0.681	0.824 / 0.985
Anastassiadis	178	0.959	0.891	0.696	0.945 / 0.965
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.80$–0.96), supporting their classification alongside $R(k)$. The CATDS proxy instead correlates more strongly with entropy and Gini than with the ratio on every dataset, supporting the statement in Methods that CATDS behaves like a member of the distribution-based family.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

Table S3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.417	0.952	0.456
entropy(literal)	0.417	1.000	0.424	0.336
Gini	0.952	0.424	1.000	0.343
ratio	0.456	0.336	0.343	1.000

Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.570	0.722	0.223
entropy(literal)	0.570	1.000	0.688	0.789
Gini	0.722	0.688	1.000	0.616
ratio	0.223	0.789	0.616	1.000

Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.291	0.821	0.193
entropy(literal)	0.291	1.000	0.385	0.758
Gini	0.821	0.385	1.000	0.155
ratio	0.193	0.758	0.155	1.000

Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.881	0.345	0.489
entropy	0.881	1.000	0.631	0.526
Gini(native)	0.345	0.631	1.000	0.314
ratio	0.489	0.526	0.314	1.000

On Klaeger and Metz, literal entropy correlates more strongly with the ratio than with either S-score or Gini – under this operationalization, entropy would not be classified in the distribution-based family at all on either of these two datasets. On Davis, literal entropy decouples from both families (consistent with Figure 1 of the main text). The native-Gini check on Anastassiadis

preserves the pattern only partly: the native Gini correlates with the entropy (0.63) but weakly and about equally with the S-score (0.35) and the ratio (0.31).

Table S4: Candidate measure cross-dataset validation

Table S4 reports three checks for each dataset. Orientation is the correlation with the number of active kinases, where more negative means more correctly oriented toward "fewer active targets = more selective". D3 is reported as in Methods, as the percentage of pairs of scored compounds whose order reverses somewhere in $B = [x_0, \tau^*]$. It is given for the emphasis parameter β with the gate x_0 fixed (the featured measure), for the gate x_0 itself swept as a free parameter (the like-for-like comparison with the entropy's baseline, Figure 6 of the main text), and for the entropy's own baseline. D4 is the percentage of compounds for which the score does not rise when one kinase is added at the reliability threshold τ^*, the strongest binding D4 allows and so the addition most likely to raise the score. B is $[5.0, 6.0]$ on Davis and Klaeger, $[4.0, 6.0]$ on Metz, and $[20, 50]$% inhibition on Anastassiadis, and the smoothing width is $T = \tau^* - x_0$ on every dataset (Methods). D3 passes at $\varepsilon = 1\%$. (`candidate_validation.py`, `d3_pair_reversals.py`)

Table S4: Candidate measure: orientation, D3 (pairs reversing in B, %), and D4, across all four datasets. Orientation and D3 are computed over the compounds that D1 allows to be scored (68, 206, 134, and 699).

Dataset	n_{cpd}	orientation r	D3, gate fixed	D3, gate swept	D3, entropy	D4 (%)
Davis	68	−0.969	0.48	8.52	8.82	100.0
Klaeger	222	−0.951	0.54	10.57	11.20	100.0
Anastassiadis	178	−0.936	0.43	12.49	10.27	100.0
Metz	704	−0.783	0.72	22.45	22.60	100.0

The "gate swept" and "entropy" columns confirm, on every dataset, that the candidate's D3 satisfaction rests on fixing the gate x_0 by declaration. Swept over the same interval, the candidate's gate reverses about as many pairs as the entropy's baseline does, and the emphasis parameter, with the gate fixed, reverses fewer than one pair in a hundred. Removing the gate inverts the measure's orientation entirely (gated $r = -0.951$ vs. ungated $r = +0.930$ against n_{active} on Klaeger), which is why the gate is required for correct orientation.

S5: Variants of the candidate measure

The featured candidate is one member of a family of measures. Three axes of variation are reported here: the floor, the Rényi order, and the smoothing width T. Each variant is checked against D1–D4 as in the main text. D4 is proved only for the featured measure (Shannon order, $\beta = x_0$, $T \geq 0.58\,(\tau^* - x_0)$), and is otherwise checked by adding one kinase at τ^* to every compound. (`family_construction.py`, `family_panelsize.py`, `candidate_validation.py`)

Quantile floor. Replacing the fixed floor x_0 with a per-compound low quantile of the profile (the 10th percentile) gives a variant that is invariant to shifts of the readout. On Davis and Klaeger it ranks compounds exactly as the featured measure does (Spearman 1.000), no pair of scored compounds reverses as the quantile is varied from the 5th to the 15th percentile, and D4 holds for every compound.

Rényi order. Generalizing the Shannon functional to Rényi order $\alpha > 0$,

$$-H_\alpha(p) = \frac{1}{\alpha - 1} \log_2 \sum_i p_i^\alpha,$$

gives a family that recovers the featured measure at $\alpha = 1$. D1 holds for every order, since the gate is applied before the entropy is computed, and D2 holds vacuously. D4 holds for additions at or below the floor, because $p_i^\alpha = 0$ whenever $p_i = 0$, and holds on every compound with one kinase added at τ^* for each order tested. D3 separates the orders (Table S5): the orders 0.5 and 1 pass on both datasets, $\alpha = 2$ is borderline (it passes on Davis and narrowly fails on Klaeger), and $\alpha = 3$ fails on both.

Table S5: Rényi-order variants on Davis and Klaeger: D3 (pairs reversing in B, %), agreement with the Shannon ranking (Spearman), and panel size p^* on Klaeger (smallest panel reaching mean $\rho > 0.90$). On Metz every order also reaches $p^* = 80$. (`family_construction.py`, `family_panelsize.py`)

Order	D3, Davis	D3, Klaeger	Spearman vs. $\alpha = 1$	p^*, Klaeger
$\alpha = 0.5$	0.61	0.13	0.999–1.000	80
$\alpha = 1$	0.48	0.54	—	80
$\alpha = 2$	0.92	1.09	0.999	80
$\alpha = 3$	1.80	1.68	0.997–0.998	80

The order does not affect panel recoverability. Every order tested, from 0.01 to 3, reaches $p^* = 80$ on both Klaeger and Metz, and the mean convergence curves differ by at most 0.003 between orders at every panel size. We feature $\alpha = 1$ because it is the Shannon entropy, for which D4 is proved.

Smoothing width. The featured measure uses $T = \tau^* - x_0$. On Klaeger, over $T = 0.5$, 0.75, 1, 1.5, and 2 (in units of $\tau^* - x_0 = 1$), orientation is unchanged ($r \approx -0.95$) and the rankings agree closely across all five values (worst-case pairwise Spearman 0.999). D3 passes for $T \geq 0.75$ (pairs reversing as the emphasis baseline is swept: 0.93%, 0.54%, 0.17%, 0.04% for $T = 0.75$, 1, 1.5, 2) and narrowly fails at $T = 0.5$ (1.94%), which also falls below the D4 bound $T \geq 0.58$. As $T \to 0$ the softplus converges to the hard cutoff, recovering the entropy together with its D3 and D4 failures.

Table S6: Provenance of reported results

Every result in the main text is printed by a script in the code repository. Table S6 gives, for each result, the script that computes it. Each script saves its printed output as `outputs/<script>.txt`; numbers in parentheses are numbered sections of that output, and a few results are read from the CSV file the script writes, named in parentheses. Random steps are seeded, so rerunning a script reproduces its output.

Table S6: Provenance of reported results, grouped by manuscript section.

Location	Result	Script (output section)
Methods, Datasets; Candidate	Active fractions; stricter S-score cutoff on Davis; missing entries and noise floor on Anastassiadis; Metz coverage per compound	`dataset_descriptive_stats.py`
Methods, Datasets	Active fractions of the 46 compounds shared by Davis and Klaeger	`claim_checks.py` (9)
Methods, Datasets	Klaeger confidence filter: rows removed, relaxed-filter sensitivity	`klaeger_filter_sensitivity.py`
Methods, Datasets	Metz coverage distribution	`metz_coverage.py`
Methods, Datasets	Metz rankings vs number of measured kinases	`claim_checks.py` (10)
Methods, Definitions	Compounds with no kinase above the threshold; ratio off-target cap	`dataset_descriptive_stats.py`
Methods, Ratio; Discussion	Bosc et al. scores and CATDS proxy (Table S2)	`bosc_catds_comparison.py`
Methods, Faithfulness; Figure 1	Published-form entropy and Gini (Table S3); operationalization scatter	`original_operationalization_robustness.py`, `metric_fidelity_robustness.py`
Methods, Rank Stability	Table 1 with mean instead of median ranks	`claim_checks.py` (3)
Methods, Candidate	Gate and smoothing-width bound for D4, including every β in B	`d4_entropy_vs_candidate.py` (1)
Table 1	Correlations between definitions (Davis, Klaeger)	`selectivity_analysis.py`, `klaeger_analysis.py`
Results, Two Families	Klaeger correlations without zero-active compounds; Gini bootstrap	`claim_checks.py` (2)
Figure 2; Results, Replication	Correlations on four datasets; replicated instability relations	`additional_datasets_analysis.py` (`cross_dataset_summary.csv`)
Results, Replication	Within-family and ratio ranges, all and active-only compounds	`claim_checks.py` (11)
Results, Replication; D1	Zero-active contrast within each definition's sweep	`claim_checks.py` (13)
Results, Replication; Type 1; D1	Pooled zero-active contrast, with and without undefined configurations; where each definition ranks zero-active compounds	`claim_checks.py` (20)
Table 2; Figure 3	Profile features vs within-family instability	`instability_by_family.py`
Results, Instability	Overall rank instability, mean and range	`klaeger_analysis.py`, `selectivity_analysis.py`
Results, Instability	Entropy instability vs share of kinases crossing the baseline sweep	`claim_checks.py` (8)

Location	Result	Script (output section)
Results, Instability	Range vs instability holding the number of active kinases fixed	`claim_checks.py` (15)
Results, Instability	Skewness association and threshold weight	`claim_checks.py` (16)
Results, Instability	Range vs S-score instability on the 46 shared compounds	`claim_checks.py` (17)
Results, Instability	Table 2 with a Bonferroni correction	`claim_checks.py` (21)
Results, Instability	Anastassiadis gap–ratio relation without saturated compounds	`claim_checks.py` (4)
Results, Instability	Undefined skewness and range: placeholder sensitivity	`feature_placeholder_sensitivity.py`
Results, Instability	Further candidate features	`instability_predictor_candidates.py`
Results, Sources	The 20 most unstable Klaeger compounds by type	`claim_checks.py` (6)
Results, Types 1 and 3	Instability and median ranks of individual Klaeger compounds	`klaeger_analysis.py` (`klaeger_selectivity_results.csv`)
Results, Type 1	Were the zero-active compounds' targets quantified	`claim_checks.py` (1)
Results, Type 3	Ten largest entropy–ratio disagreements	`claim_checks.py` (5)
Results, Type 3	Instability by number of active kinases; most stable compounds	`claim_checks.py` (7, 18)
Figure 4	Binding profiles of three Davis compounds	`binding_profiles.py`
Figure 5; Table S1	Panel-size recovery and p^* at four bars	`panel_size_analysis.py`, `pstar_bar_sensitivity.py`
D1; Figure 6	Zero-active instability under each convention; D1 and D3 panels	`undefined_score_convention.py`, `failure_mode_illustration.py`
Table 3; D2	Ratio at near-tied top targets	`d2_near_tie_sensitivity.py`
Table 3; D3	Pairs reversing in B	`d3_pair_reversals.py`
Candidate; Table S4	Orientation, D3, D4 at τ^*, smoothing-width sweep	`candidate_validation.py`
Candidate; Figure 7	Agreement with the entropy; panel-size recovery, gated and ungated	`candidate_measure.py`
Candidate	Gini p^* without compounds whose Gini is undefined	`claim_checks.py` (14)
Candidate; Section S5	Variants of the candidate and their panel sizes	`family_construction.py`, `family_panelsize.py`
Smooth Hinge	Entropy and candidate under one added kinase at τ^*; worked example	`d4_entropy_vs_candidate.py` (2–6)
Smooth Hinge	Realistic panel expansion	`claim_checks.py` (19)
Discussion	Platform vs definition on the 46 shared compounds	`davis_klaeger_overlap.py`, `platform_vs_definition.py`
Discussion	Instability vs cross-platform rank change	`claim_checks.py` (9)
Discussion	Maximum affinities of the zero-active compounds	`failure_mode_illustration.py`
Discussion	Compounds shared by Klaeger and Anastassiadis	`claim_checks.py` (12)

Location	Result	Script (output section)
Discussion, Limitations	Clinical outcomes	`clinical_outcomes_correlation.py`, `faers_pull.py`